

ROBUST VOICE ACTIVITY DETECTION USING A MASKED AUDITORY ENCODER BASED CONVOLUTIONAL NEURAL NETWORK

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ABSTRACT

Voice activity detection (VAD) based on deep learning has achieved remarkable success. However, when the traditional features (e.g., raw waveforms and MFCCs) are directly fed to the deep neural network model, the performance decreases because of noise interference. Here, we propose a robust VAD approach using a masked auditory encoder based convolutional neural network (M-AECNN). First, we analyze the effectiveness of using auditory features as deep learning encoder. These features can roughly simulate the transmission of sound to human inner-ear hair cells; thus, they are more robust than the raw waveform and frequency domain features designed as encoders. Second, similar to the human ear's masking effect for different speech frequencies, the proposed auditory encoder can further improve the robustness of VAD by increasing the gain for cleaner speech frequencies. Extensive experimental results demonstrate that this approach achieves about 10.5% absolute improvement in the area under the curve on the AURORA-2J dataset compared with a VAD method based on a CNN and MFCCs.

Index Terms— Voice activity detection, auditory encoder, convolutional neural network

1. INTRODUCTION

Voice activity detection (VAD) is a technique to detect whether a sound signal belongs to speech or non-speech based on the statistical distribution of acoustic features. It plays an important role in front-end processing for various speech applications such as speech enhancement, robust speech recognition systems, and speaker recognition.

In recent decades, many VAD algorithms have been proposed [1, 2, 3, 4]. Most previous methods are effective for clean environments. However, there are many types of noise in the real world, and these noise sources will seriously interfere with the performance of VAD tasks. Therefore, a robust VAD algorithm is needed in practical applications.

Recently, VAD based on deep learning (DL) has shown significant improvements in robustness because of its strong classification capability. Examples include VAD based on deep neural networks (DNNs) [5, 6, 7, 8], convolutional neural networks (CNNs) [9], and long short-term memory networks (LSTMs) [10, 11]. The general approach is to view VAD as a multi-label classification problem, where the non-linear classification function is parametrized by the different types of acoustic features used as input. We also can refer to input acoustic features as the encoder and classification as decoder [12]. However, direct use of these DL systems in speech/non-speech classification often cannot fully mine speech information under noisy conditions, which reduces VAD robustness. The most important reason for the above problem is that, the encoders of the network are not robust, that is, they are designed to use energy levels, zero-crossings, log-spectrograms, Mel-frequency cepstral coefficients (MFCCs), and raw waveforms [6, 10, 13].

Generally, low-dimensional features extracted from the first layer are a prerequisite for high-dimensional feature information learning [14]. Therefore, designing a neural network encoder or the input feature is important for DL-based VAD and other speech tasks. Motivated by the fact that human auditory system, which can easily distinguish between speech and non-speech by sensing the intensity and frequency of speech and noise, we propose a masked auditory encoder (AE) based CNN (M-AECNN) to improve the robustness of DL-based VAD. Our main contributions are summarized as follows:

1) To simulate the sound processing of the cochlea, the gammachirp auditory filterbank (GAFB) [15] is used to split noisy speech into a different subband speech signal. In addition, many previous studies have shown that modulated signals can be used to simulate the transmission of sound signals to inner-ear hair cells [16, 17]. In this paper, we explore use the GAFB and a modulated speech signal to simulate this process, and directly use it as the encoder for the CNN to improve the robustness of VAD.

2) The response of the human auditory system to different

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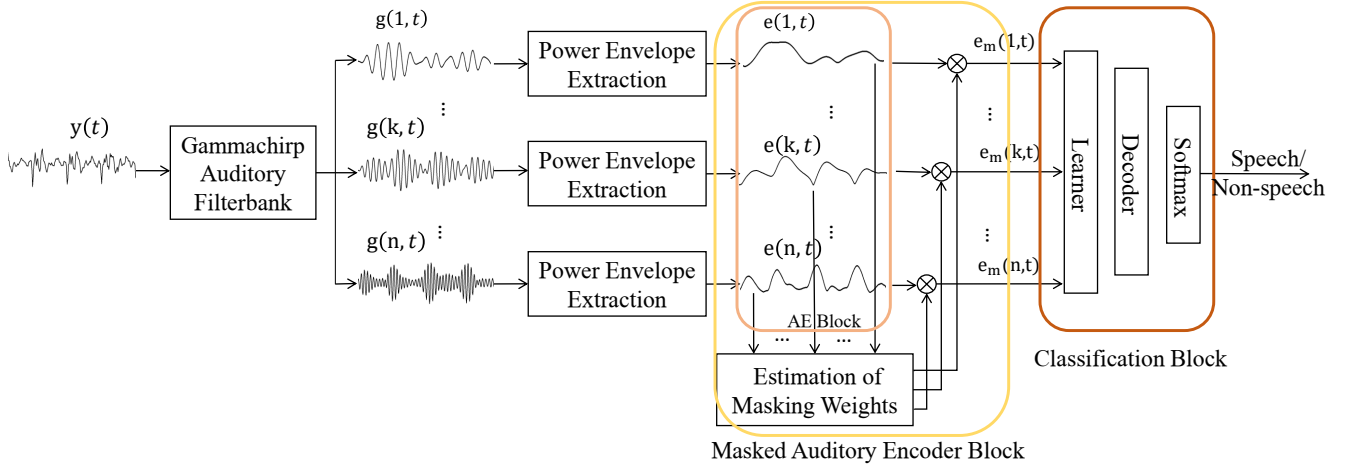


Fig. 1. The flow chart of the proposed method.

frequencies of sound differs; when listening to a signal with speech at one frequency, noise signals at other frequencies are suppressed [18, 19]. Based on the above mechanism, M-AECNN is further proposed to manage the AE gain control. In the M-AECNN, an estimation of masking weights block is proposed to give a higher weight to the cleaner subbands, which represent different frequency bands.

2. MASKED AUDITORY ENCODER BASED CONVOLUTIONAL NEURAL NETWORK

Recently, many studies have shown that using CNNs to learn the target task directly from a raw waveform [10] designed encoder can achieve better results than using frequency-domain features. However, a raw speech waveform is subjected to strong fluctuations caused by noise, this will cause the statistical distributions of speech and noise to overlap because they can have similar statistical properties. Motivated by the mechanism of the reception of speech signals in humans, we wanted to see if there was a robust way to learn the auditory feature directly in the CNN as an encoder. An overview of the proposed VAD method is shown in Fig. 1.

2.1. Auditory encoder

The AE comprises two parts: an auditory filterbank and a module to calculate the modulated signal. In the auditory filterbank, noisy speech $y(t)$ is first split into different subbands by the GAFB. This filterbank can reproduce the psychophysiological estimation of human auditory filters over a wide range of central frequencies. The output of the k -th subband signal is represented by

$$g(k, t) = g_c(k, t) * y(t), 1 < k < n, \quad (1)$$

where $g(k, t)$ is the k -th subband of noisy speech, t is the index of the time domain sample, k is the index for an auditory

filterbank subband, $g_c(k, t)$ is the impulse response of the k -th subband, and $*$ is the convolution operation. The following formula obtains the impulse response:

$$g_c(k, t) = At^{a-1} \exp(-2\pi w_f \text{ERB_N}(f_k) t) \times \cos(2\pi f_k t + c \ln(t) + \varphi), \quad (2)$$

where $At^{a-1} \exp(-2\pi w_f \text{ERB_N}(f_k) t)$ is the amplitude term represented by a gamma distribution, A is the amplitude, and a and w_f are the filter order and bandwidth of the filter, respectively. Moreover, c is a parameter for the frequency modulation or the chirp rate, and φ is the original phase. In addition, $\ln(t)$ is a natural logarithm of time, f_k is the center frequency of the k -th filter, and ERB_N is the equivalent rectangular bandwidth, which is calculated as follows:

$$\text{ERB_N}(f_k) = \frac{f_k}{Q_{\text{ear}}} + B_{\text{min}}, \quad (3)$$

where f_k is the central frequency of the k -th subband. Moreover, Q_{ear} is the asymptotic filter quality at high frequencies, and B_{min} is minimum bandwidth; they are respectively set as 9.26 and 24.7.

To calculate the modulated signal, we use the Hilbert transform to calculate the subband modulation power envelope. The k -th subband modulation power envelope is

$$e(k, t) = |g(k, t) + j \text{Hilbert}(g(k, t))|. \quad (4)$$

Lastly, the AE of CNN can be denoted as $\mathbf{e} \in \mathbb{R}^{F \times D \times N}$, F is the number of context frame, D is frame length and N is the subband number.

2.2. Masked auditory encoder based CNN

As shown in Fig.2, our basic framework is based on 2D CNN, which can be an excellent approach for learning the temporal relationship of noisy speech contexts. However, simply using

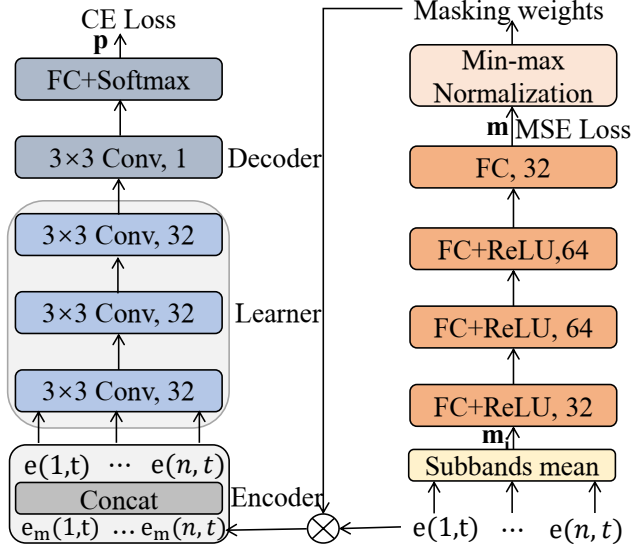


Fig. 2. Proposed M-AECNN. The right block is the masking weights estimation.

the original noisy AE as the CNN encoder does not lead to VAD with sufficient robustness. To simulate the masking effect of human ears on different frequencies, we propose estimation of masking weights block to learn a group of masking values for AE.

Estimation of masking weights: In the proposed structure for estimation of masking weights, we first compute the input $\mathbf{m}_i = [m_{i,1}, \dots, m_{i,k}, \dots, m_{i,n}]$, where the elements are the means of the speech subbands

$$m_{i,k} = \frac{1}{D} \sum_{t=1}^D e(k, t), \quad (5)$$

where D is the length of the frame. We use a neural network composed of four full connection (FC) layers, and maximum-minimum normalization to obtain weights on each subband. The masking weights $\mathbf{w}_m = [w_{m,1}, \dots, w_{m,k}, \dots, w_{m,n}]$ and masked subband $\mathbf{e}_m$ are formulated as

$$\mathbf{w}_m = \frac{\mathbf{m} - \min(\mathbf{m})}{\max(\mathbf{m}) - \min(\mathbf{m})}, \quad (6)$$

$$\mathbf{e}_m = \mathbf{e}^T \times \mathbf{w}_m, \quad (7)$$

where $\mathbf{m} = [m_1, \dots, m_k, \dots, m_n]$ is the output of the last FC layer. Normalization is used to limit $\mathbf{m}$ to $[0, 1]$. The hidden units of FCs are $[32, 64, 64, 32]$ respectively. To avoid losing information, the input multiple subband auditory encoder and masked encoder are concatenated for each subband as $\mathbf{e}_i = [\mathbf{e}, \mathbf{e}_m] \in \mathbb{R}^{F \times D \times 2N}$.

Using a CNN with three 2D convolutional layers with a kernel size of 3×3 and 32 filters, we can learn the encoder vector to high dimensional. After each layer in the

CNN, there is mean-pooling with a $[1, 2, 3, 1]$ shape to remove redundant information. Finally, a 2D convolutional layer with a single filter is used as decoder. An FC layer with a Softmax activation function is used to obtain the final speech/non-speech probability $\mathbf{p} = [p_1, p_2]$.

Loss function: The loss function used by the proposed VAD method in the training stage is cross-entropy (CE), which is calculated from the following function

$$\text{CE}(\mathbf{p}) = -[x_1 \cdot \log(p_1) + x_2 \cdot \log(p_2)], \quad (8)$$

where $\mathbf{x} = [x_1, x_2]$ represents the one-hot coding of the ideal state, which is speech "[0,1]" or non-speech "[1,0]". Moreover, $\mathbf{p}$ is the probability that the predicted speech frame is correct. To estimate the masking weights, the mean squared error loss (MSE) is used

$$\text{MSE}(\mathbf{m}) = \frac{1}{n} \sum_{k=1}^n (m_k - z_k)^2, \quad (9)$$

where z_k is the average of the k -th subband auditory feature of clean speech. The total loss of the network in the training stage is

$$L_{\text{total}} = \text{CE}(\mathbf{p}) + \lambda \text{MSE}(\mathbf{m}). \quad (10)$$

During the testing stage, it is

$$\text{Output} = \begin{cases} \text{speech} & , p_1 < p_2 \\ \text{non-speech} & , p_1 \geq p_2 \end{cases} \quad (11)$$

3. EXPERIMENTS

3.1. Data descriptions

To perform the final evaluation, we used the speech data of the AURORA-2J [20] and NOISEX-92 [21]. In the training stage, 8,440 clean speech utterances of AURORA-2J were selected to create the clean data. White, pink, factory, and babble noise from NOISEX-92 were then used to artificially create the noisy speech signals. The final audio sample rate was resampled at a rate of 16KHz. Finally, we use a GAFB with 32 filters to divide the signal into different subbands. Noisy speech signals with signal-to-noise ratios (SNRs) of 20, 15, 10, 5, 0, -5, and -10 were generated. We chose 7,740 utterances as the training set and 700 utterances as the cross-validation set. The test set was the test set of AURORA-2J, which has 1,001 utterances, with noise added at the corresponding SNR. The noise types for each utterance were the same as those of the training data and were randomly selected.

3.2. Experimental setup

The structure of the DNN used in this study has the same configuration. It comprises three hidden layers, each containing 512 cells. The LSTM model has three hidden layers and 512 units in each hidden layer. The CNN comprises four 2D

Table 1. Comparison of the areas under the curve (AUCs) (%) obtained using different DL structures.

Architecture	Encoder	20 dB	15 dB	10 dB	5 dB	0 dB	-5 dB	-10 dB	Average
DNN	MFCC	92.1	90.1	87.7	84.9	82.2	78.4	68.8	83.5
	STFT	91.8	88.9	85.2	82.3	80.9	79.5	72.9	83.1
LSTM	MFCC	91.9	90.7	88.7	86.1	82.7	78.5	69.0	83.9
	STFT	91.1	90.0	87.7	85.6	82.0	77.9	70.8	83.6
CNN	MFCC	91.6	90.5	88.3	85.6	82.2	77.4	68.6	83.5
	STFT	91.9	90.3	88.7	87.2	85.1	81.2	71.7	85.1
	RW	93.4	93.3	93.2	92.2	90.3	85.6	76.9	89.3
AECNN	AE_0	93.9	93.9	93.6	93.0	92.3	88.9	79.4	90.7
	AE	95.2	95.0	94.8	94.5	94.0	91.3	83.9	92.7
M-AECNN	AE	96.3	96.2	95.9	95.3	94.6	92.7	87.1	94.0

convolutional layers, and the number of channels in each convolutional layer was set to 32, 32, 32, and 1, respectively. The convolution kernels of the 2D CNN have a shape of 3×3 . The activation functions of the DNN, LSTM, and CNN structures in the hidden layers were all ReLU, and their loss function was the most basic CE loss function. In practice, the learning rate for all methods in this study was 0.01, and the batch size was 64.

As for the MFCC-based VAD method, 13-dimensional MFCC features were extracted from each frame of noisy speech, where the frame length was 400 and the frameshift was 200. In this study, a short-time Fourier transform (STFT) was used for the log-spectrogram feature. Moreover, its frame length and frameshift are the same as those of MFCC. The raw waveform (RW) of noisy speech was also divided into frames with a frame length of 400 samples and a frameshift of 200. AE_0 is the RW that passes through GAFB; it is the AE without the power envelope extraction. The frame length and frameshift of AE are the same as those of RW. To enhance the context information of the input feature, each frame of the feature used in this study was combined with two frames in each context to form the final neural network input feature. The hyperparameter λ was set to 100 in this study.

The area under the curve (AUC) is the area under the receiver operating characteristic curve in relation to a coordinate axis in which the ordinate and abscissa are the false acceptance rate (FAR) and false rejection rate (FRR) like [22].

3.3. Results and analysis

As the results show Table 1, VAD methods based on DL and frequency-domain features (MFCC and log-spectrogram) have limited performance and are easily affected by noise. RW is the original waveform based encoder; it also represent the direct use of the original waveform bypassing the first layer of CNN. The VAD methods based on the RW average achieve an absolute improvement of 5.8% over the CNN

based on MFCCs.

The result of AE_0 is better than that of RW. Because speech and noise are represented differently under different subbands, we can use CNN to mine the acoustic information of these subbands. These results also demonstrate that using an AE is effective for a CNN-based VAD method. Comparing the frequency and time domains, the AE obtains better results on the VAD task. This means that designing an encoder by simulating the process of sound to inner hair cell is beneficial to CNN-based VAD. If the proposed M-AECNN is used to improve the weights of cleaner frequency bands, better VAD performance can be obtained than if the original CNN is used. The proposed approach average achieves an absolute improvement in AUC of about 10.5% over the CNN-MFCC based approach; in the -5 dB and -10 dB low-SNR noisy environments, this improvement more obvious. The results of this study demonstrate the potential significance of using human auditory properties for robust VAD.

4. CONCLUSION AND FUTURE WORK

In this paper, we proposed a robust VAD method that uses a masked auditory encoder-based CNN. Our proposed VAD method performs substantially better than the DNN-, LSTM-, and CNN-based robust VAD methods in the frequency and time domains. In future work, we will also apply the idea of the proposed robust VAD method to acoustic models.

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